pip install pandas ## to insatl pandas

```
import pandas as pd

#series
#Pandas Series is a one dimensioanl array like objects
data = [1,2,3,4,5]
series = pd.Series(data)
print("Series:\n",series)
print(type(series))
```

```
Series:
 0    1
1    2
2    3
3    4
4    5
dtype: int64
<class 'pandas.Series'>
```

```
## create a series from dictionary
data = {'a':1,'b':2,'c':3}
series_dict = pd.Series(data)
print(series_dict)
```

```
a    1
b    2
c    3
dtype: int64
```

```
data = [10,20,30]
indeX = ['a','b','c']
pd.Series(data, index = indeX)
```

```
a    10
b    20
c    30
dtype: int64
```

```
#data frame
data = {
    'Name': ['Krish','Jack','Hughi'],
    "Age" : [10,20,30],
    "City" : ["bangalore",'Mysore','New York']
}

df = pd.DataFrame(data)
print(df)
print(type(df))
```

```
    Name  Age       City
0  Krish   10  bangalore
1   Jack   20     Mysore
2  Hughi   30   New York
<class 'pandas.DataFrame'>

df = pd.read_csv("D:\Skill Lab\Skill Lab 3rd Sem\sales_data.csv")
df.head(5)

   Transaction ID        Date Product Category              Product
Name  \
0           10001  2024-01-01      Electronics            iPhone 14
Pro
1           10002  2024-01-02  Home Appliances         Dyson V11
Vacuum
2           10003  2024-01-03         Clothing         Levi's 501
Jeans
3           10004  2024-01-04            Books        The Da Vinci
Code
4           10005  2024-01-05  Beauty Products  Neutrogena Skincare
Set

   Units Sold  Unit Price  Total Revenue         Region Payment Method

0           2      999.99        1999.98  North America    Credit Card

1           1      499.99         499.99         Europe         PayPal

2           3       69.99         209.97           Asia     Debit Card

3           4       15.99          63.96  North America    Credit Card

4           1       89.99          89.99         Europe         PayPal


df.tail(5)

     Transaction ID        Date Product Category  \
235           10236  2024-08-23  Home Appliances
236           10237  2024-08-24         Clothing
237           10238  2024-08-25            Books
238           10239  2024-08-26  Beauty Products
239           10240  2024-08-27           Sports

                                       Product Name  Units Sold  Unit
Price  \
235  Nespresso Vertuo Next Coffee and Espresso Maker           1
159.99
236                        Nike Air Force 1 Sneakers           3
90.00
237           The Handmaid's Tale by Margaret Atwood           3
```

```
10.99
238              Sunday Riley Luna Sleeping Night Oil            1
55.00
239                         Yeti Rambler 20 oz Tumbler            2
29.99

     Total Revenue         Region Payment Method
235         159.99         Europe         PayPal
236         270.00           Asia     Debit Card
237          32.97  North America    Credit Card
238          55.00         Europe         PayPal
239          59.98           Asia    Credit Card
```

```
df['Product Name']
```

```
0                                       iPhone 14 Pro
1                                    Dyson V11 Vacuum
2                                    Levi's 501 Jeans
3                                   The Da Vinci Code
4                              Neutrogena Skincare Set
                            ...
235    Nespresso Vertuo Next Coffee and Espresso Maker
236                           Nike Air Force 1 Sneakers
237              The Handmaid's Tale by Margaret Atwood
238                Sunday Riley Luna Sleeping Night Oil
239                           Yeti Rambler 20 oz Tumbler
Name: Product Name, Length: 240, dtype: str
```

```
df.loc[0,'Product Name']  ## loc is used to locate info from csv file
```

```
'iPhone 14 Pro'
```

```
df.loc[0,:]    ## list slice to get all columns at index 0
```

```
Transaction ID              10001
Date                   2024-01-01
Product Category      Electronics
Product Name        iPhone 14 Pro
Units Sold                      2
Unit Price                 999.99
Total Revenue             1999.98
Region              North America
Payment Method        Credit Card
Name: 0, dtype: object
```

```
df.loc[[0,1,2], :]
```

```
   Transaction ID        Date Product Category      Product Name
Units Sold  \
0           10001  2024-01-01      Electronics     iPhone 14 Pro
2
```

```
1           10002  2024-01-02  Home Appliances  Dyson V11 Vacuum
1
2           10003  2024-01-03         Clothing  Levi's 501 Jeans
3

   Unit Price  Total Revenue         Region Payment Method
0      999.99        1999.98  North America    Credit Card
1      499.99         499.99         Europe         PayPal
2       69.99         209.97           Asia     Debit Card
```

```
df.loc[0:2,:]    ## in loc value after colon is included
```

```
   Transaction ID        Date Product Category      Product Name
Units Sold  \
0           10001  2024-01-01      Electronics     iPhone 14 Pro
2
1           10002  2024-01-02  Home Appliances  Dyson V11 Vacuum
1
2           10003  2024-01-03         Clothing  Levi's 501 Jeans
3

   Unit Price  Total Revenue         Region Payment Method
0      999.99        1999.98  North America    Credit Card
1      499.99         499.99         Europe         PayPal
2       69.99         209.97           Asia     Debit Card
```

```
df.loc[0:2]    ## without columns prints evvery columns
```

```
   Transaction ID        Date Product Category      Product Name
Units Sold  \
0           10001  2024-01-01      Electronics     iPhone 14 Pro
2
1           10002  2024-01-02  Home Appliances  Dyson V11 Vacuum
1
2           10003  2024-01-03         Clothing  Levi's 501 Jeans
3

   Unit Price  Total Revenue         Region Payment Method
0      999.99        1999.98  North America    Credit Card
1      499.99         499.99         Europe         PayPal
2       69.99         209.97           Asia     Debit Card
```

```
df.loc[0:2, ['Product Name',"Units Sold"]]  ## Only given columns will
be printed
```

```
       Product Name  Units Sold
0     iPhone 14 Pro           2
1  Dyson V11 Vacuum           1
2  Levi's 501 Jeans           3
```

```
df[['Product Name',"Units Sold"]].head(3)
```

```
       Product Name  Units Sold
0     iPhone 14 Pro           2
1  Dyson V11 Vacuum           1
2  Levi's 501 Jeans           3
```

```
df.iloc[[0,7],[0,3]]    ##Wee can print middle elements and random
elements
```

```
   Transaction ID          Product Name
0           10001         iPhone 14 Pro
7           10008  Blueair Classic 480i
```

```
df.iloc[0:2,0:4]
```

```
   Transaction ID        Date Product Category      Product Name
0           10001  2024-01-01      Electronics     iPhone 14 Pro
1           10002  2024-01-02  Home Appliances  Dyson V11 Vacuum
```

```
df.iloc[0:2,:]
```

```
   Transaction ID        Date Product Category      Product Name
Units Sold  \
0           10001  2024-01-01      Electronics     iPhone 14 Pro
2
1           10002  2024-01-02  Home Appliances  Dyson V11 Vacuum
1

   Unit Price  Total Revenue         Region Payment Method
0      999.99        1999.98  North America    Credit Card
1      499.99         499.99         Europe         PayPal
```

```
## acessing a specified element using iat
df.iat[2,2]
```

```
'Clothing'
```

```
#data frame
data = {
    'Name': ['Krish','Jack','Hughi'],
    "Age" : [10,20,30],
    "City" : ["bangalore",'Mysore','New York']
}

df = pd.DataFrame(data)
print(df)
print(type(df))
```

```
    Name  Age       City
0  Krish   10  bangalore
1   Jack   20     Mysore
2  Hughi   30   New York
<class 'pandas.DataFrame'>
```

```
## Adding a column
df['Salary']=[50000,60000,70000]
df
```

```
    Name  Age       City  Salary
0  Krish   10  bangalore   50000
1   Jack   20     Mysore   60000
2  Hughi   30   New York   70000
```

```
print(df.drop('Salary',axis = 1))
print()
print(df)    ## without inplace = True original data remains same
```

```
    Name  Age       City
0  Krish   10  bangalore
1   Jack   20     Mysore
2  Hughi   30   New York

    Name  Age       City  Salary
0  Krish   10  bangalore   50000
1   Jack   20     Mysore   60000
2  Hughi   30   New York   70000
```

```
df
```

```
    Name  Age       City  Salary
0  Krish   10  bangalore   50000
1   Jack   20     Mysore   60000
2  Hughi   30   New York   70000
```

```
df.drop('Salary',axis=1,inplace=True)

df
```

```
    Name  Age       City
0  Krish   10  bangalore
1   Jack   20     Mysore
2  Hughi   30   New York
```

```
df['Age']  = df['Age']+2
df
```

```
    Name  Age       City
0  Krish   12  bangalore
1   Jack   22     Mysore
2  Hughi   32   New York
```

```
df.drop(0,inplace=True)  ## by default inplace is false  ## by default
axis value is zero
df
```

```
    Name  Age      City
1   Jack   22    Mysore
2  Hughi   32  New York
```

```
df = pd.read_csv("D:\Skill Lab\Skill Lab 3rd Sem\sales_data.csv")
df.head(5)
```

```
   Transaction ID        Date Product Category             Product
Name  \
0           10001  2024-01-01      Electronics            iPhone 14
Pro
1           10002  2024-01-02  Home Appliances         Dyson V11
Vacuum
2           10003  2024-01-03         Clothing         Levi's 501
Jeans
3           10004  2024-01-04            Books        The Da Vinci
Code
4           10005  2024-01-05  Beauty Products  Neutrogena Skincare
Set

   Units Sold  Unit Price  Total Revenue         Region Payment Method

0           2      999.99        1999.98  North America    Credit Card

1           1      499.99         499.99         Europe         PayPal

2           3       69.99         209.97           Asia     Debit Card

3           4       15.99          63.96  North America    Credit Card

4           1       89.99          89.99         Europe         PayPal
```

```
# Display the data type of each col
print("Data types: \n",df.dtypes)
print()
#  Describe the data Frame
print("Statistical Summary : \n",df.describe())
```

```
Data types:
 Transaction ID        int64
Date                    str
Product Category        str
Product Name            str
Units Sold            int64
Unit Price          float64
Total Revenue       float64
Region                  str
Payment Method          str
dtype: object
```

```
Statistical Summary :
       Transaction ID  Units Sold   Unit Price  Total Revenue
count       240.00000  240.000000   240.000000     240.000000
mean      10120.50000    2.158333   236.395583     335.699375
std          69.42622    1.322454   429.446695     485.804469
min       10001.00000    1.000000     6.500000       6.500000
25%       10060.75000    1.000000    29.500000      62.965000
50%       10120.50000    2.000000    89.990000     179.970000
75%       10180.25000    3.000000   249.990000     399.225000
max       10240.00000   10.000000  3899.990000    3899.990000
```

```
df = pd.read_csv("data.csv")
## fetch the first five rows
df.head(5)
```

```
         Date Category  Value   Product  Sales Region
0  2023-01-01        A   28.0  Product1  754.0   East
1  2023-01-02        B   39.0  Product3  110.0  North
2  2023-01-03        C   32.0  Product2  398.0   East
3  2023-01-04        B    8.0  Product1  522.0   East
4  2023-01-05        B   26.0  Product3  869.0  North
```

```
df.tail(5)
```

```
          Date Category  Value   Product  Sales Region
45  2023-02-15        B   99.0  Product2  599.0   West
46  2023-02-16        B    6.0  Product1  938.0  South
47  2023-02-17        B   69.0  Product3  143.0   West
48  2023-02-18        C   65.0  Product3  182.0  North
49  2023-02-19        C   11.0  Product3  708.0  North
```

```
df.describe()
```

```
           Value       Sales
count  47.000000   46.000000
mean   51.744681  557.130435
std    29.050532  274.598584
min     2.000000  108.000000
25%    27.500000  339.000000
50%    54.000000  591.500000
75%    70.000000  767.500000
max    99.000000  992.000000
```

```
df.dtypes
```

```
Date            str
Category        str
Value       float64
Product         str
Sales       float64
```

```
Region          str
dtype: object
```

```
df.isnull().any()
```

```
Date        False
Category    False
Value        True
Product     False
Sales        True
Region      False
dtype: bool
```

Home Work: Revise and come SQL topic 1)groupBy 2)Joining

```
df.isnull().sum()   ## it is giving the counts  of null values present
```

```
Date        0
Category    0
Value       3
Product     0
Sales       4
Region      0
dtype: int64
```

```
df_filled = df.fillna(0)
df_filled
```

	Date	Category	Value	Product	Sales	Region
0	2023-01-01	A	28.0	Product1	754.0	East
1	2023-01-02	B	39.0	Product3	110.0	North
2	2023-01-03	C	32.0	Product2	398.0	East
3	2023-01-04	B	8.0	Product1	522.0	East
4	2023-01-05	B	26.0	Product3	869.0	North
5	2023-01-06	B	54.0	Product3	192.0	West
6	2023-01-07	A	16.0	Product1	936.0	East
7	2023-01-08	C	89.0	Product1	488.0	West
8	2023-01-09	C	37.0	Product3	772.0	West
9	2023-01-10	A	22.0	Product2	834.0	West
10	2023-01-11	B	7.0	Product1	842.0	North
11	2023-01-12	B	60.0	Product2	0.0	West
12	2023-01-13	A	70.0	Product3	628.0	South
13	2023-01-14	A	69.0	Product1	423.0	East
14	2023-01-15	A	47.0	Product2	893.0	West
15	2023-01-16	C	0.0	Product1	895.0	North
16	2023-01-17	C	93.0	Product2	511.0	South
17	2023-01-18	C	0.0	Product1	108.0	West
18	2023-01-19	A	31.0	Product2	578.0	West
19	2023-01-20	A	59.0	Product1	736.0	East
20	2023-01-21	C	82.0	Product3	606.0	South

```
21  2023-01-22        C   37.0  Product2  992.0  South
22  2023-01-23        B   62.0  Product3  942.0  North
23  2023-01-24        C   92.0  Product2  342.0   West
24  2023-01-25        A   24.0  Product2  458.0   East
25  2023-01-26        C   95.0  Product1  584.0   West
26  2023-01-27        C   71.0  Product2  619.0  North
27  2023-01-28        C   56.0  Product2  224.0  North
28  2023-01-29        B    0.0  Product3  617.0  North
29  2023-01-30        C   51.0  Product2  737.0  South
30  2023-01-31        B   50.0  Product3  735.0   West
31  2023-02-01        A   17.0  Product2  189.0   West
32  2023-02-02        B   63.0  Product3  338.0  South
33  2023-02-03        C   27.0  Product3    0.0   East
34  2023-02-04        C   70.0  Product3  669.0   West
35  2023-02-05        B   60.0  Product2    0.0   West
36  2023-02-06        C   36.0  Product3  177.0   East
37  2023-02-07        C    2.0  Product1    0.0  North
38  2023-02-08        C   94.0  Product1  408.0  South
39  2023-02-09        A   62.0  Product1  155.0   West
40  2023-02-10        B   15.0  Product1  578.0   East
41  2023-02-11        C   97.0  Product1  256.0   East
42  2023-02-12        A   93.0  Product3  164.0   West
43  2023-02-13        A   43.0  Product3  949.0   East
44  2023-02-14        A   96.0  Product3  830.0   East
45  2023-02-15        B   99.0  Product2  599.0   West
46  2023-02-16        B    6.0  Product1  938.0  South
47  2023-02-17        B   69.0  Product3  143.0   West
48  2023-02-18        C   65.0  Product3  182.0  North
49  2023-02-19        C   11.0  Product3  708.0  North
```

```
## filling missing values with the mean of the column
df['Sales'].mean()
```

```
np.float64(557.1304347826087)
```

```
df['Sales_FillNA'] = df["Sales"].fillna(df['Sales'].mean())
df.head(5)
```

```
         Date Category  Value   Product  Sales Region  Sales_FillNA
0  2023-01-01        A   28.0  Product1  754.0   East         754.0
1  2023-01-02        B   39.0  Product3  110.0  North         110.0
2  2023-01-03        C   32.0  Product2  398.0   East         398.0
3  2023-01-04        B    8.0  Product1  522.0   East         522.0
4  2023-01-05        B   26.0  Product3  869.0  North         869.0
```